

Shihan Fu

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Research Interest

My research focuses on [human-centered AI](#).
I aim to understand how people use technologies within their own knowledge, tasks, and situational contexts, and then design intelligent systems that can adapt to these human factors.

Education

Northeastern University

PhD in Computer Science

- Advised by Prof. Varun Mishra

Sep 2024 - present

The Hong Kong University of Science and Technology

Mphil in Computational Media and Arts

- Advised by Prof. Mingming Fan

Sep 2022 - Jul 2024

Zhejiang University

B.A. in Russian Language and Literature

Sep 2017 - June 2022

Work Experience

Amazon - Applied Scientist Intern

Search Experience Science Group

- Developed and deployed a testbed version of an LLM-based conversational shopping assistant on the WebArena simulated shopping environment.
- Designed and evaluated conversation strategies grounded in HCI theories of decision making and conversational interaction.

Jul 2025 - Nov 2025

Publications

Under Review

- [P.1] **"It Felt Like I Was Left in the Dark": Exploring Information Needs and Design Opportunities for Family Caregivers of Older Adult Patients in Critical Care Settings**

[Shihan Fu](#), Bingsheng Yao, Smit Desai, Yuqi Hu, Yuling Sun, Samantha Stonbraker, Yanjun Gao, Elizabeth M Goldberg, Dakuo Wang

Link: <https://arxiv.org/abs/2502.05115>

- [P.2] **LLM Agent Meets Agentic AI: Can LLM Agents Simulate Customers to Evaluate Agentic-AI-based Shopping Assistants?**

Lu Sun, [Shihan Fu](#), Bingsheng Yao, Yuxuan Lu, Wenbo Li, Hansu Gu, Jiri Gesi, Jing Huang, Chen Luo, Dakuo Wang

Link: <https://arxiv.org/abs/2509.21501>

- [P.3] **Human and LLM-based voice assistant interaction: An analytical framework for user verbal and nonverbal behaviors**

Szeyi Chan, [Shihan Fu](#), Jiachen Li, Bingsheng Yao, Smit Desai, Mirjana Prpa, Dakuo Wang

Link: <https://arxiv.org/abs/2408.16465>

- [P.4] **Clinical Challenges and AI Opportunities in Decision-Making for Cancer Treatment-Induced Cardiotoxicity**
Siyi Wu, Weidan Cao, **Shihan Fu**, Bingsheng Yao, Ziqi Yang, Changchang Yin, Varun Mishra, Daniel Addison, Ping Zhang, Dakuo Wang
Link: <https://arxiv.org/abs/2408.03586>

CONFERENCE PAPERS (PEER-REVIEWED)

- [C.7] **CardioAI: A Multimodal AI-based System to Support Symptom Monitoring and Risk Prediction of Cancer Treatment-Induced Cardiotoxicity**

Siyi Wu*, Weidan Cao*, **Shihan Fu**, Bingsheng Yao, Ziqi Yang, Changchang Yin, Varun Mishra, Daniel Addison, Ping Zhang, Dakuo Wang

Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems, 2025

- [C.6] **SepsisCalc: Integrating Clinical Calculators into Early Sepsis Prediction via Dynamic Temporal Graph Construction**

Changchang Yin, **Shihan Fu**, Bingsheng Yao, Thai-Hoang Pham, Weidan Cao, Dakuo Wang, Jeffrey M Caterino, Ping Zhang

KDD (Knowledge Discovery and Data Mining) 2025 ADS Track, 2025 (In press)

- [C.5] **Bridging the Literacy Gap for Adults: Streaming and Engaging in Adult Literacy Education through Livestreaming**

Shihan Fu, Jianhao Chen, Emily Kuang, Mingming Fan

Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems, 2024

DOI: 10.1145/3613904.3642423

-  [C.4] **"It is hard to remove from my eye": Design Makeup Residue Visualization System for Chinese Traditional Opera (Xiqu) Performers**

Zeyu Xiong, **Shihan Fu**, Yanying Zhu, Chenqing Zhu, Xiaojuan Ma, Mingming Fan

Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems, 2024

DOI: 10.1145/3613904.3642261 (Honorable Mention (Top 5%))

- [C.3] **To Reach the Unreachable: Exploring the Potential of VR Hand Redirection for Upper Limb Rehabilitation**

Peixuan Xiong, Yukai Zhang, Nandi Zhang, **Shihan Fu**, Xin Li, Yadan Zheng, Jinni Zhou, Xiquan Hu, Mingming Fan

Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems, 2024

DOI: 10.1145/3613904.3642912

-  [C.2] **OperArtistry: An AR-based Interactive Application to Assist the Learning of Chinese Traditional Opera (Xiqu) Makeup**

Zeyu Xiong*, **Shihan Fu***, Mingming Fan

Proceedings of the Eleventh International Symposium of Chinese CHI, 2023

DOI: 10.1145/3629606.3629622 (Honorable Mention (Top 5%))

- [C.1] **LOOP Meditation: Enhancing Novice's VR Meditation Experience with Physical Movement**

Shihan Fu, Liangliang Qiang, Wei Zeng

Proceedings of the 16th International Symposium on Visual Information Communication and Interaction, 2023

DOI:10.1145/3615522.3615538

EXTENDED ABSTRACTS & WORKSHOP PAPERS (PEER-REVIEWED)

[P.1] **AromaBlendz: An Olfactory System for Crafting Personalized Scents**

Shihan Fu, Jianhao Chen, Yi Cai, Mingming Fan

Extended Abstracts of the CHI Conference on Human Factors in Computing Systems, 2024

DOI: 10.1145/3613905.3648670

Work in Progress

Designing wearable and voice assistant technologies to support family caregivers of older adults in managing stress

- Supported by the NIH AI-CARING Institute (National Artificial Intelligence Institute for Aging and Intelligent Care). <https://www.ai-caring.org/>
- Demo video showcasing the system's stress monitoring and conversational support features

[Watch here](#)

Honors and Awards

2022-2024 **Postgraduate Studentship (PGS)
Full Scholarship**

Service

SERVICE TO THE PROFESSION

2026	Reviewer CHI 2026, JMIR
2025	Reviewer CHI 2025, CSCW 2025
2024	Reviewer IMWUT 2024, IJHCS(International Journal of Human-Computer Studies)

Skills

Design and Research

User Research, Data Analysis, Prototyping, Usability Testing
User Experience (UX) and User Interface (UI) Design
Storyboarding, Journey Mapping, Personas, and Scenarios